

Toward Guidance-Free AR Visual Generation via CCA

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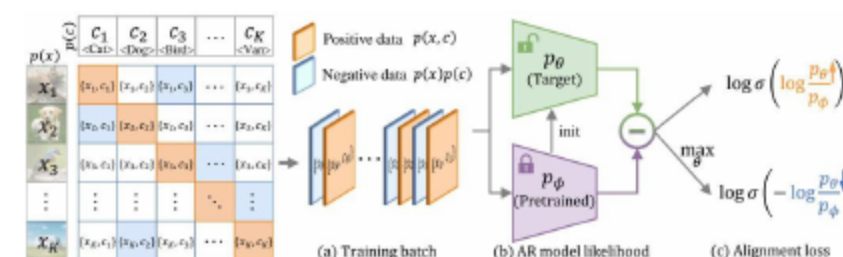
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Motivation & Overview

3 Introduction: Challenges & Approach Overview

- Tokenized-image AR seeks unified vision-language modeling.
- Yet AR models rely on CFG, breaking this unification.
- CFG doubles sampling cost and adds instability, tuning.
 - Guidance dependence creates design mismatches and burden.
 - CCA aligns to the desired sampling distribution; no sampler change.
- Delivers guidance-free sampling; quality kept; bridges methods.

4 Background: AR, Guidance, and Alignment



- AR models decode images as sequences of tokens.
 - Visual tokenizers quantize pixels; often kept frozen.
 - CFG shifts sampling logits, boosting fidelity.
- Guidance needs two model passes per token step.
 - LM DPO aligns by contrasting preferred vs rejected.
 - Inspires CCA: contrastive alignment for AR tokens.

Method

Condition Contrastive Alignment

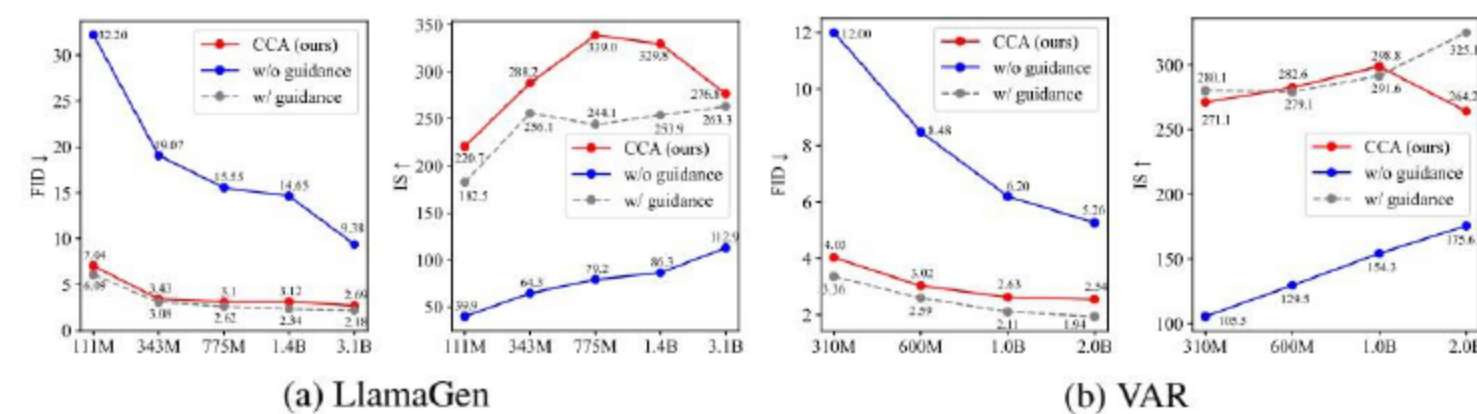
6 CCA: Derivation and Practical Implementation

- Contrast condition-consistent vs. inconsistent samples
- Align model scores via NCE to the target distribution
- Parameterize conditional residuals via pretrained+target
 - No test-time guidance; λ controls fidelity-diversity
 - Use positive pairs and shuffled negatives in each batch
- Fine-tune one AR model; no extra unconditional model; β tunes

7

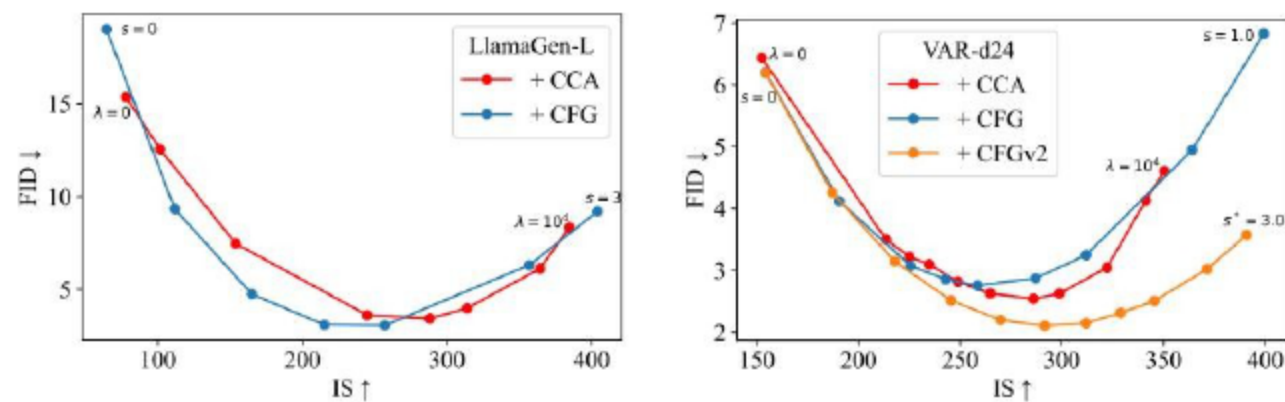
Experiments and Results

↑ Effectiveness of CCA Across Sizes



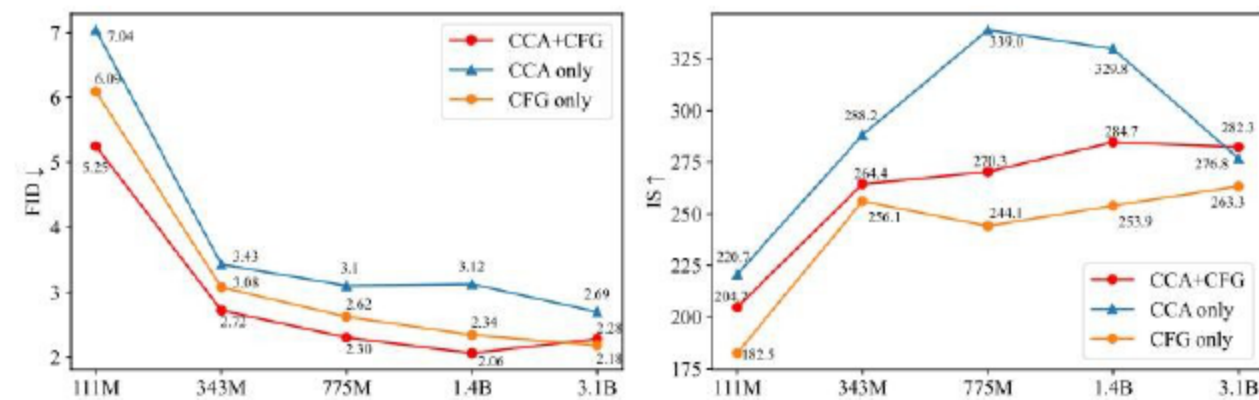
- Consistent FID ↓ and IS ↑ across LlamaGen and VAR
 - Matches CFG quality while avoiding guidance cost
 - Unified sampling, no extra guidance passes needed
- Improvements hold from 100M to multi-B params

λ Diversity-Fidelity: CCA vs CFGv2



- FID-IS curves: CCA matches or exceeds CFG/CFGv2.
 - λ in CCA yields smooth, monotonic trade-off control.
 - Balances diversity and fidelity without altering sampling.
- Directly aligns conditional residuals to target distribution.

1 CCA+CFG: Synergy and Scaling



- CCA+CFG outperforms CFG alone across model sizes.
 - Shows positive scaling: gains persist from 111M to 3.1B.
 - Requires distinct λ and s tuning when combined with CFG.
- CCA aligns AR vision natively; DPO-style methods lag.

11 Qualitative: Diversity and Fidelity



- CCA yields sharper textures and clearer edges.
 - Category consistency improves across prompts.
 - Stable diversity maintained without guidance.
- Qualitative IS gains align with FID trends.
 - Comparable to CFG with lower sampling overhead.
 - Extra collages provided as backup qualitative.

12 Conclusion and Key Takeaways

- CCA aligns AR generators during training, guidance-free.
- Improves FID/IS and bridges gaps between model types.
- Single knob λ tunes fidelity-diversity trade-offs.
 - Unifies language alignment and visual guidance methods.
 - Composes with CFG when needed; lowers operational costs.
- Future: broader data, richer signals, deeper theory links.